

PROJECT WORK

ELEVATE LABS

INTERNSHIP

**TITLE: GLOBAL CO2 EMISSIONS
TRACKER BY SECTOR**

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OBJECTIVE: Build a dashboard to track Carbon Emissions from energy, transport, and industry sector across countries.

TOOLS: Tableau/ Excel/ Python (for data prep)

MINI- GUIDE:

Import multi-year emissions dataset.

Prepare per capita and per GDP metrics.

Use Tableau maps and bar graphs by sector.

DELIVERABLES:

Global Emissions dashboard.

PDF policy brief on top polluters.

Executive Summary

The global fight against climate change requires urgent action to curb carbon dioxide (CO₂) emissions. This brief identifies the leading country and sectoral polluters, analyses emission trends, and recommends policy actions to drive emissions reductions across energy, transport, and industry sectors.

Key Findings

- **Top Emitting Countries (2024 Estimates):**

1. China – 29% of global CO₂ emissions
2. United States – 14%
3. India – 7%
4. European Union (27 countries) – 6%
5. Russia – 5%

- **Emissions by Sector Globally:**

Energy (Electricity and Heat Production): ~40%

Industry (Manufacturing, Cement, Chemicals): ~20%

Transport (Road, Aviation, Shipping): ~15%

- **Per Capita Emissions (Highest):**

Qatar, United States, Australia among the highest emitters per capita.

- **CO₂ Emissions per GDP Unit (Carbon Intensity):**

Emerging economies show higher emissions intensity relative to GDP compared to developed economies.

Causes and Impacts

- Energy Sector: Heavy dependence on coal and fossil fuels, especially in China and India.
- Transport Sector: Rising vehicle ownership, inefficient fuel standards, and rapid aviation growth.
- Industry Sector: Steel, cement, and chemical production remain energy-intensive and carbon-heavy.
- Impact: Rising temperatures, extreme weather events, biodiversity loss, and socio-economic vulnerabilities, especially for developing nations.

Policy Recommendations

1. Accelerate Renewable Energy Transition

- Subsidize solar, wind, and green hydrogen development.
- Retire coal-fired power plants early.

2. Enhance Transport Efficiency

- Invest in electric vehicle infrastructure.
- Implement strict vehicle emissions standards.

3. Decarbonize Industry

- Incentivize carbon capture and storage (CCS) technologies.
- Promote low-carbon manufacturing processes.

4. Strengthen International Cooperation

- Align national emissions targets with the Paris Agreement.
- Support climate finance for developing nations.

Conclusion

Reducing global CO₂ emissions demands bold policy interventions targeting the energy, transport, and industry sectors. Prioritizing these efforts can steer the world towards a low-carbon future and help mitigate the devastating impacts of climate change.

References

- Global Carbon Project (2024)
- International Energy Agency (IEA) Emissions Report 2024
- UNFCCC Climate Data Portal

Thank You!!